

Group A2: The EVO-ECO Matrix: Simulating Cellular Plasticity as a Markov Decision Process"

If a cancer cell is an "Agent" trying to maximize its survival during therapy, there must be a mathematical trade-off. Using Reinforcement Learning (a Markov Decision Process), **at what specific "metabolic/fitness cost" does reversible phenotypic plasticity become a worse survival strategy than simply proliferating and waiting for a random genetic mutation?**

Your Deliverable (2 Weeks): You must build a "Toy" Reinforcement Learning simulation to prove your answer.

- **Biologists:** You are responsible for defining the "Rules of the Game." You must define the States (e.g., stem vs. differentiated), Actions, and biologically accurate Reward Functions (the cost of surviving vs. the energy cost of dedifferentiation).
- **Computational Students:** You are responsible for building the RL Engine. You must translate the biological rules into a Python-based Q-learning algorithm that simulates the cell's survival policy over 1,000 generations of adaptive therapy.

Submit your findings in the standard Module Report format, including the code, the biological justification for your Reward Function, and a graph showing the "tipping point" where plasticity fails.